

How Risky Are Denied Mortgage Applicants? Partial Identification Bounds from Linked Administrative Records

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June 24, 2026

Abstract

We estimate bounds on the counterfactual default rate of denied mortgage applicants (what would have happened if they had been approved) and characterize the *breakdown frontier*: the maximum degree of adverse selection under which each policy conclusion holds. This selective labels problem arises because default outcomes are observed only for originated loans. We link the universe of U.S. mortgage applications from HMDA (2018–2022, excluding 2019) with Freddie Mac loan performance data through a cell-level stochastic linkage covering 11,136 risk cells. Under monotone selection and a bounded adverse selection (κ -shift) assumption, the width of the identified set equals κ by construction, so we treat κ as a sensitivity parameter and map how policy conclusions vary across its range. Our central finding is that policy conclusions are robust to substantial adverse selection: for any $\kappa \leq 0.10$ and transport gap $\delta_S \leq 0.01$, the minimax-optimal lending policy still approves more borrowers than the status quo, implying that, under our benchmark welfare parameters, current denial rates can be rationalized only if denied applicants are at least 10 percentage points riskier than observably similar approved borrowers. As a descriptive benchmark, the observed default gradient across denial-rate quintiles is only 2.2 percentage points (0.9% to 3.1%)—roughly five times smaller than the adverse selection required to justify current denials. Under our welfare objective and sample composition, the binding constraint on policy conclusions is transport uncertainty, the gap between Freddie Mac-observed and population default rates, rather than the degree of adverse selection.

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Keywords: Partial identification, breakdown frontier, selective labels, mortgage default, counterfactual risk, administrative data linkage, robust policy learning

JEL Classification: G21, C14, D81

1 Introduction

Mortgage lending involves a fundamental selection problem: we observe repayment outcomes only for applicants whose loans were approved. The roughly 8% of conventional mortgage applicants who are denied each year remain counterfactual: we do not know what their default behavior would have been had they received loans. Yet understanding this counterfactual is essential for evaluating lending standards, calibrating credit risk models, and designing regulatory policy.

This paper provides the first large-scale partial identification analysis of the denied-group counterfactual default rate in the U.S. mortgage market. We link two administrative datasets: the Home Mortgage Disclosure Act (HMDA) data, which covers the near-universe of mortgage applications with approval/denial outcomes, and the Freddie Mac Single-Family Loan-Level Dataset (SFLLD), which provides post-origination performance data for a substantial subset of originated loans. Because individual-level linkage is not possible between these datasets, we construct a cell-level stochastic linkage based on year, state, loan amount, debt-to-income ratio, and combined loan-to-value ratio, yielding 11,136 risk cells (excluding 2019) with sufficient data for estimation.

Our identification strategy builds three layers of assumptions, each progressively tightening the bounds on the denied-group counterfactual default rate $E[Y(1)|D = 0, X \in c]$:

1. **Manski bounds** (no assumptions): $p_R(c) \in [0, 1]$.
2. **Monotone selection**: denied applicants are weakly riskier than approved applicants within each cell, so $p_R(c) \geq p_A(c)$.
3. **κ -shift**: adverse selection is bounded, $p_R(c) \leq p_A(c) + \kappa$, yielding $p_R(c) \in [\hat{p}(c), \hat{p}(c) + \kappa]$.

Because Freddie Mac covers only approximately 40% of originated loans, we introduce a transport layer parameterized by δ_S that accounts for the potential gap between the Freddie-observed default rate $\hat{p}(c)$ and the population originated default rate $p_A(c)$.

Our main specification (excluding 2019 due to COVID forbearance contamination of performance measures) yields denied-weighted aggregate bounds of $[0.023, 0.123]$ at $\kappa = 0.10$. The width of the identified set equals κ by construction, a transparent relationship between assumptions and conclusions. In the spirit of the breakdown frontier approach of [Masten & Poirier \(2020\)](#), we treat κ as a sensitivity parameter and ask: *how much adverse selection is needed before each policy conclusion reverses?* This reframes the paper’s contribution from the precision of the bounds to the robustness of the conclusions drawn under them.

We document three empirical findings. First, for any $\kappa \leq 0.10$ and transport gap $\delta_S \leq 0.01$, the minimax-optimal cell-level lending policy approves more borrowers than

the status quo (96.4% versus 93.8% at $\delta_S = 0$; 94.9% versus 93.8% at $\delta_S = 0.01$). Under our benchmark welfare parameters ($r = 0.02$, $\text{LGD} = 0.35$, $\text{FDC} = 0.005$), current denial patterns can be rationalized under worst-case partial identification only if denied applicants are at least 10 percentage points riskier than approved borrowers in the same risk cell. As a descriptive benchmark, the observed Freddie default rate rises from 0.9% in the lowest-denial-rate quintile to 3.1% in the highest, a 2.2 percentage point gradient that is roughly five times smaller than the κ required to justify current denials.

Second, we conduct a two-parameter breakdown analysis over (κ, δ_S) , where δ_S parameterizes the Freddie-to-population default rate gap. The “expand lending” conclusion is robust across the entire κ range we consider (0–0.20) so long as transport uncertainty remains moderate ($\delta_S \leq 0.01$). At $\delta_S = 0.02$, welfare falls by 56% and the optimal policy begins restricting lending below status quo levels. Under our welfare objective and sample composition, transport uncertainty is a more consequential sensitivity dimension than adverse selection, a constructive finding, because the Freddie-to-population gap is, in principle, resolvable with additional data linkage.

Third, we develop a minimax-robust policy learning framework that optimizes cell-level lending decisions under worst-case default risk. The LP relaxation yields nearly binary solutions, making the framework computationally tractable and its policy prescriptions sharp.

Related literature. This paper connects to several strands of the literature. The selective labels problem in algorithmic decision-making has received growing attention (Lakkaraju et al., 2017; Kleinberg et al., 2018; Arnold et al., 2022), typically in the context of pretrial bail or hiring decisions. We contribute to this literature by applying partial identification methods to a large-scale credit market setting with linked administrative data.

Our identification approach builds on the partial identification framework of Manski (1990, 2003) and the monotone selection models of Lee (2009). Our breakdown-style sensitivity analysis is motivated by Masten & Poirier (2020), who develop formal inference methods for the frontier of assumptions under which a conclusion holds; we apply their conceptual framing to a two-parameter sensitivity space (κ, δ_S) in a large-scale empirical setting, evaluated over a grid of parameter values. Our confidence intervals follow Imbens & Manski (2004).

The policy learning component relates to the statistical treatment rules literature (Manski, 2004; Stoye, 2009; Kitagawa & Tetenov, 2018), extended to a setting with partial identification and linear programming.

Finally, our paper contributes to the mortgage default literature that uses HMDA and Freddie Mac data (Bhutta & Hizmo, 2021; Mian & Sufi, 2009; Keys et al., 2010), by explicitly

addressing the selection problem that arises from observing outcomes only for originated loans.

The remainder of the paper is organized as follows. Section 2 presents the identification framework. Section 3 describes the data and cell construction. Section 4 presents the main empirical results, including the breakdown analysis. Section 5 develops the policy learning framework. Section 6 concludes.

2 Identification Framework

2.1 Setup and Target Estimand

Consider a population of mortgage applicants indexed by observable characteristics X that define risk cells c . Let $D \in \{0, 1\}$ denote the lending decision ($D = 1$: originated, $D = 0$: denied), and let $Y(1) \in \{0, 1\}$ denote the potential default outcome under origination.

Our target estimand is the *denied-group counterfactual default rate*:

$$p_R(c) \equiv E[Y(1) \mid D = 0, X \in c], \quad (1)$$

the average default rate that denied applicants in cell c would have experienced had they been approved.

We observe:

- From HMDA: the number of originated (n_{orig}) and denied (n_{denied}) applicants in each cell c , along with the origination rate $\pi(c) = n_{\text{orig}} / (n_{\text{orig}} + n_{\text{denied}})$.
- From Freddie Mac: the realized default rate among a subset $S \subseteq \{D = 1\}$ of originated loans, $\hat{p}(c) = E[Y \mid S = 1, X \in c]$, with coverage ratio $s(c) = n_{\text{Freddie}} / n_{\text{orig}}$.

Let $p_A(c) = E[Y(1) \mid D = 1, X \in c]$ denote the true default rate among all originated loans. The key challenge is that $\hat{p}(c)$ proxies for $p_A(c)$ but may not equal it, since Freddie Mac purchases are not a random sample of originations.

2.2 Layer 0: Manski Bounds

Without any assumptions, the counterfactual default rate is bounded only by the logical constraint $p_R(c) \in [0, 1]$.

2.3 Layer 1: Monotone Selection

Assumption 1 (Monotone Selection). *Within each cell c , denied applicants are weakly riskier than approved applicants: $p_R(c) \geq p_A(c)$.*

This assumption reflects the economic logic of lending: conditional on observables X , lenders use private information (credit reports, employment verification, etc.) to screen applicants, approving those with lower perceived risk. Under Assumption 1 and perfect transport ($p_A = \hat{p}$), the lower bound tightens to $\hat{p}(c)$.

2.4 Layer 2: κ -Shift Bounds

Assumption 2 (κ -Shift). *Adverse selection is bounded: $p_R(c) \leq p_A(c) + \kappa$ for a known constant $\kappa \geq 0$.*

The parameter κ controls the maximum degree of adverse selection within a cell. Under Assumptions 1–2 and perfect transport ($\hat{p} = p_A$):

$$p_R(c) \in [\hat{p}(c), \min(1, \hat{p}(c) + \kappa)]. \quad (2)$$

The width of the identified set is exactly κ (when $\hat{p} + \kappa \leq 1$). Rather than viewing this as a limitation, we treat κ as a transparent sensitivity parameter and characterize the values of κ (and δ_S ; see Section 2.5) at which specific policy conclusions reverse. We report results for $\kappa \in \{0, 0.05, 0.10, 0.20\}$.

2.5 Transport Layer

Because Freddie Mac covers only a subset of originated loans, $\hat{p}(c)$ may differ from $p_A(c)$.

Assumption 3 (Transport Band). *$|p_A(c) - \hat{p}(c)| \leq \delta_S$ for a known constant $\delta_S \geq 0$.*

Under all three assumptions, the combined bounds become:

$$p_R(c) \in [\max(0, \hat{p}(c) - \delta_S), \min(1, \hat{p}(c) + \delta_S + \kappa)]. \quad (3)$$

We calibrate δ_S empirically using within-Freddie origination channel variation (Section 4.5.1).

2.6 Aggregate Bounds

For the aggregate denied-group counterfactual, we report denied-weighted means:

$$E[Y(1) \mid D = 0] = \sum_c \frac{n_{\text{denied}}(c)}{N_{\text{denied}}} \cdot p_R(c), \quad (4)$$

where the weight reflects our target population (denied applicants).

2.7 Inference

We construct 95% confidence intervals using the [Imbens & Manski \(2004\)](#) method for partially identified parameters. Standard errors are computed via a two-stage bootstrap ($B = 500$): we resample state×year clusters with replacement, then within each resampled cell add a parametric perturbation $\hat{p}_b(c) = \hat{p}(c) + \varepsilon$, $\varepsilon \sim N(0, \sqrt{\hat{p}(1 - \hat{p})/n_{\text{Freddie}}})$, to account for within-cell sampling noise.

3 Data and Sample Construction

3.1 HMDA

The Home Mortgage Disclosure Act requires most mortgage lenders to report data on applications and originations. We use HMDA Loan Application Register (LAR) data for 2018–2022, restricting to conventional purchase mortgages for first-lien, owner-occupied, site-built, single-unit properties. We retain applications with action taken $\in \{1$ (originated), 3 (denied) $\}$.

Table 1 reports summary statistics for this initial sample, which comprises approximately 14.3 million applications, of which 13.1 million (92%) were originated and 1.1 million (8%) were denied. Mean loan amounts range from \$298,000 (2018) to \$407,000 (2022), reflecting house price appreciation. The conforming loan amount restriction is applied during cell construction (Section 3.3), yielding the analysis sample reported in Table 3.

Exclusion of 2019. Our main specification excludes the 2019 origination cohort. The 24-month ever-90-days-past-due rate for 2019 Freddie Mac originations is 4.66%, far exceeding the 1.3–2.8% range for other cohorts. This anomaly reflects COVID-19 forbearance reporting: borrowers who entered forbearance programs in 2020–2021 were frequently reported as 90+ days delinquent despite not experiencing true credit losses. The 2019 cohort’s 24-month performance window (through early 2021) falls squarely within the forbearance period. We report 2019 results separately as a robustness check.

3.2 Freddie Mac SFLLD

The Freddie Mac Single-Family Loan-Level Dataset provides monthly performance data for loans purchased by Freddie Mac. We compute three default indicators: (i) `ever_90plus_24m`: whether the loan was ever 90+ days delinquent within 24 months (our primary measure); (ii) `ever_90plus_36m`: the 36-month analog; and (iii) `severe_default_24m`: foreclosure, short sale or charge-off, or REO disposition within 24 months (zero-balance codes 02, 03, 09).

Table 2 reports Freddie Mac summary statistics. The sample comprises 4.6 million loans. Mean credit scores are approximately 755, mean DTI is 35%, and mean CLTV is 83%. The primary default rate (`ever_90plus_24m`) averages 1.3–2.8% for non-2019 cohorts and 4.66% for 2019. Severe default rates are near zero across all cohorts.

3.3 Cell-Level Stochastic Linkage

Because HMDA and Freddie Mac cannot be linked at the individual level, we construct a cell-level stochastic linkage. We first restrict the HMDA sample to loans classified as Conforming (C) or Undetermined (U) in the HMDA `conforming_loan_limit` field, excluding loans explicitly flagged as Nonconforming (NC).¹ Cells are defined by:

$$c = (\text{year}) \times (\text{state}) \times (\text{loan amount bin}) \times (\text{DTI bin}) \times (\text{CLTV bin}).$$

This yields 33,098 unique cells across all years. We retain cells with at least 30 Freddie Mac loans for reliable default rate estimation, leaving 13,745 cells (11,136 excluding 2019) covering 92.8% of HMDA applications but only 74.8% of denied applications. The coverage gap arises because omitted cells, those with few or no Freddie Mac matches, have substantially higher denial rates (27.0% versus 6.2% for included cells), reflecting that high-denial cells tend to have sparse Freddie Mac coverage (e.g., cells with unusual loan amount, DTI, or CLTV combinations where Freddie purchases few loans).

Freddie Mac coverage. Among included cells, the median Freddie coverage ratio is $s = 0.40$ (mean $s = 0.50$), meaning Freddie Mac loans typically constitute 40–50% of all originations. This imperfect coverage motivates our transport layer (Section 2.5).

¹We retain Undetermined (U) classifications because excluding them would drop a non-trivial share of applications whose conforming status is ambiguous due to reporting variation, without clear evidence that these loans differ systematically from those flagged Conforming.

4 Empirical Results

4.1 Main Bounds

Table 3 reports denied-group counterfactual bounds for the main specification (ex-2019). The denied-weighted observed Freddie default rate is $\hat{p} = 2.31\%$. Under monotone selection alone, this provides a lower bound with no finite upper bound (short of 100%).

Introducing the κ -shift assumption yields:

- $\kappa = 0.05$: $[0.023, 0.073]$, width = 5pp
- $\kappa = 0.10$: $[0.023, 0.123]$, width = 10pp
- $\kappa = 0.20$: $[0.023, 0.223]$, width = 20pp

At our benchmark $\kappa = 0.10$, the 95% Imbens-Manski confidence interval is $[0.021, 0.126]$, only slightly wider than the identified set, indicating that sampling uncertainty is small relative to identification uncertainty. Figure 1 visualizes the identified set across κ values.

The year-by-year decomposition (Table 3) reveals notable temporal variation: $\hat{p}$ ranges from 1.5% (2021, low interest rate environment) to 3.7% (2018). Even after excluding 2019, the 2018 cohort exhibits meaningfully higher default rates than 2020–2022 cohorts.

Full-sample outer bounds. Across all years (including 2019), our included cells cover 74.8% of denied applicants. For the remaining 25.2% in omitted cells, we impose only Manski bounds $[0, 1]$. The resulting all-year full-sample outer bounds at $\kappa = 0.10$ are $[0.022, 0.349]$ (Table 3, last row), substantially wider, driven entirely by the Manski contribution from omitted cells.

4.2 Breakdown Analysis

Because the width of the identified set equals κ by construction, we shift focus from the bounds themselves to a breakdown-style sensitivity analysis: for what values of (κ, δ_S) does each policy conclusion reverse? This follows the conceptual framing of Masten & Poirier (2020), applied here to a two-parameter sensitivity space.

To preview the policy framework developed in Section 5: a social planner maximizes worst-case welfare across cells, where welfare from approving cell c depends on the mortgage spread $r = 0.02$, loss given default $\text{LGD} = 0.35$, and false-denial cost $\text{FDC} = 0.005$. A cell is approved when

$$n_{\text{orig}}(c) \cdot [r + \text{FDC} - p_A^{\text{wc}}(c) \cdot \text{LGD}] + n_{\text{denied}}(c) \cdot [r + \text{FDC} - p_R^{\text{wc}}(c) \cdot \text{LGD}] > 0, \quad (5)$$

where $p_A^{\text{wc}} = \hat{p} + \delta_S$ and $p_R^{\text{wc}} = \hat{p} + \delta_S + \kappa$ are worst-case default rates. The cell-level threshold

for p_R^{wc} thus depends on both the welfare parameters *and* the ratio $n_{\text{orig}}/n_{\text{denied}}$: cells with many more originated than denied applicants can tolerate higher worst-case denied risk because the originated borrowers contribute positive expected welfare.

What κ is needed to justify current denials? At $\delta_S = 0$ and $\kappa = 0.10$, the minimax-optimal policy approves 96.4% of applicants, more than the status quo’s 93.8%. At $\delta_S = 0.01$, the optimal policy still approves 94.9%, above the status quo. The “expand lending” conclusion holds across the entire κ range we consider (0–0.20) so long as $\delta_S \leq 0.01$. That is, even if denied applicants’ true default rate is up to 20 percentage points higher than approved borrowers in the same cell, worst-case welfare from cell-level approval remains positive for the vast majority of cells, because the already-originated borrowers dominate the welfare calculation. Table 9 in Section 5 reports the minimax-optimal approval rate across a grid of (κ, δ_S) values.

The two-parameter sensitivity structure. The policy conclusion reverses not from increases in κ , but from increases in δ_S . At $\delta_S = 0.02$, the approval rate drops to 92.1%, below the status quo, while welfare falls by 56%. This asymmetry arises because δ_S worsens both the originated-borrower and denied-borrower terms in (5), whereas κ affects only the denied-borrower term. Since denied applicants constitute only about 6% of the included sample, κ has a mechanically smaller welfare footprint than δ_S . This structural property is nonetheless policy-relevant: it identifies transport uncertainty as the actionable margin, because the Freddie-to-population gap is resolvable with additional data linkage, whereas the true degree of adverse selection is inherently unobservable.

Comparison to observed gradients. As a descriptive plausibility benchmark, we compare the κ values in our sensitivity grid to observable variation in default rates. The denied-weighted Freddie default rate rises from 0.9% in the lowest denial-rate quintile to 3.1% in the highest, a 2.2 percentage point gradient (Section 4.3). This cross-cell gradient among approved borrowers is not a direct measure of within-cell adverse selection, since monotone selection ($p_R(c) \geq p_A(c)$) is a within-cell assumption that does not directly constrain the cross-cell relationship. Nevertheless, the 2.2pp scale provides a reference: the κ values at which policy conclusions remain robust ($\kappa = 0.10$ – 0.20) are roughly 5–10 times larger than the observed variation in approved-borrower default rates across denial-rate quintiles.

4.3 Marginal Borrower Analysis

A key economic question is whether lender denial decisions contain real information about credit risk. To address this, we group cells into quintiles by their denial rate and examine how observed default rates vary.

Table 4 reports the results, and Figure 4 plots them. The observed default rate $\hat{p}$ rises monotonically from 0.86% in Q1 to 3.05% in Q5, a 3.5-fold increase. This confirms that lender screening is informative: cells where lenders deny more frequently are indeed riskier ex post.

However, two caveats temper this finding. First, the gradient in $\hat{p}$ (2.2 percentage points from Q1 to Q5) is modest relative to the gradient in denial rates (18.4 percentage points), suggesting that incremental screening precision at the margin of approval is limited. Second, Q5 contains 40% of all denied applicants but its $\hat{p}$ (3.05%) is only marginally higher than Q4 (2.42%), indicating diminishing screening returns in the tail.

4.4 Risk Characteristics Heterogeneity

Table 5 examines heterogeneity along observable risk dimensions, and Figure 5 plots the DTI and CLTV gradients.

CLTV. The observed default rate exhibits a clear leverage gradient: $\hat{p}$ rises from 1.3% for $\text{CLTV} \leq 80\%$ to 3.1% for $\text{CLTV} > 95\%$. This $2.4\times$ increase is consistent with the well-documented relationship between leverage and default risk: borrowers with less equity have less to lose from strategic default and are more vulnerable to negative equity shocks.

DTI. Default rates increase with debt burden: $\hat{p}$ rises from 1.3% for $\text{DTI} \leq 36\%$ to 3.2% for $\text{DTI} 43\text{--}50\%$. The $\text{DTI} > 50\%$ bin is an exception ($\hat{p} = 2.9\%$), likely reflecting that these high-DTI approvals are subject to compensating factors (high assets, strong credit history) that partially offset the elevated debt burden. Notably, the $\text{DTI} > 50\%$ bin has a very high denial rate (72.5%), consistent with lenders applying intense scrutiny to this segment.

4.5 Transport Sensitivity

Table 6 reports the sensitivity of bounds to the transport parameter δ_S , which controls the assumed gap between Freddie-observed and population originated default rates. As the breakdown analysis in Section 4.2 establishes, δ_S is the binding dimension of the sensitivity space. Figure 6 shows how the identified set widens with δ_S .

Transport has a first-order effect on bound width. At $\kappa = 0.10$:

- $\delta_S = 0$: $[0.023, 0.123]$, width = 10.0pp
- $\delta_S = 0.01$: $[0.015, 0.133]$, width = 11.9pp
- $\delta_S = 0.02$: $[0.009, 0.143]$, width = 13.4pp
- $\delta_S = 0.05$: $[0.002, 0.173]$, width = 17.1pp

The lower bound is the denied-weighted aggregate of cell-level $\max(0, \hat{p}(c) - \delta_S)$, so it does not simply equal $\hat{p} - \delta_S$ at the aggregate level (the clipping at zero is applied cell by cell before aggregation).

4.5.1 Empirical Calibration of δ_S

We calibrate δ_S using within-Freddie origination channel variation. Freddie Mac loans are originated through three channels: Retail, Correspondent, and Broker. For each cell with at least two channels having ≥ 30 loans, we compute the maximum default rate gap across channels. This within-Freddie gap provides a benchmark for the scale of default rate heterogeneity within the Freddie population, though it is not a mathematical lower bound on the Freddie-to-all-originated gap since non-Freddie loans may differ along dimensions not captured by channel variation.

For the ex-2019 sample, the median within-cell channel gap is 1.4 percentage points, with the 75th percentile at 2.5pp and the 90th percentile at 4.0pp. These magnitudes support $\delta_S = 0.01$ as a conservative benchmark and $\delta_S = 0.02$ as a moderate scenario. We additionally verify that Freddie-purchased loans are broadly comparable to non-Freddie originations on HMDA observables (loan amount, income, interest rate), with differences on the order of 5–15%.

4.6 Coverage-Threshold Robustness

A natural concern is that the transport assumption is less plausible for cells where Freddie coverage is low. Table 8 restricts the sample to cells exceeding various coverage thresholds. The bounds are notably stable: $\hat{p}$ moves only from 2.3% to 2.8% as the coverage threshold increases from 0 to 0.6. This stability is reassuring: even restricting to cells where Freddie effectively covers the majority of originations (and transport is most plausible), the core finding is unchanged.

5 Policy Learning Under Partial Identification

5.1 Framework

We consider a social planner who must decide, for each cell c , whether to approve ($z_c = 1$) or deny ($z_c = 0$) all applicants. The planner faces uncertainty about $p_R(c)$ and adopts a minimax approach: maximize welfare under the worst-case realization of partially identified default rates. For the policy analysis, we additionally require cells to have at least 30 HMDA applications (in addition to the 30 Freddie Mac loans required for bounds estimation); this further filter has negligible effect on cell counts because Freddie loans are a subset of HMDA originations.

Welfare function. If cell c is approved:

$$W_c^{\text{approve}} = n_{\text{orig}}(c) \cdot [r - p_A^{\text{wc}}(c) \cdot \text{LGD}] + n_{\text{denied}}(c) \cdot [r - p_R^{\text{wc}}(c) \cdot \text{LGD}], \quad (6)$$

where $r = 0.02$ is the mortgage spread, $\text{LGD} = 0.35$ is loss given default, and the worst-case rates under transport-robust κ -shift are $p_A^{\text{wc}} = \hat{p} + \delta_S$ and $p_R^{\text{wc}} = \hat{p} + \delta_S + \kappa$.

If cell c is denied:

$$W_c^{\text{deny}} = -(n_{\text{orig}}(c) + n_{\text{denied}}(c)) \cdot \text{FDC}, \quad (7)$$

where $\text{FDC} = 0.005$ captures the social cost of false denials (deadweight loss from credit rationing).

Optimization. The planner solves:

$$\max_{z \in [0,1]^C} \sum_c z_c \cdot W_c^{\text{approve}} + (1 - z_c) \cdot W_c^{\text{deny}}, \quad (8)$$

which is a linear program solvable via standard methods (HiGHS solver).

5.2 Results

Table 7 reports policy comparison results for the ex-2019 main specification.

Status quo. The observed lending pattern, which reflects actual lender decisions, not a cell-level binary rule, yields welfare of 126,415 with an overall approval rate of 93.8%.

Unconstrained optimal. At $\kappa = 0.10$, the unconstrained minimax-optimal policy approves 96.4% of applicants and achieves welfare of 118,338. The welfare is *lower* than the status quo because the robust policy uses worst-case default rates ($\hat{p} + \kappa$), while the status quo welfare is evaluated at observed rates ($\hat{p}$). This comparison should be interpreted with caution: the status quo reflects lenders’ finer within-cell screening ability, while the robust policy is a coarser cell-level rule. They represent different policy classes and are not directly comparable.

Transport sensitivity. Transport uncertainty has a substantial welfare impact:

- $\delta_S = 0$: welfare = 118,338 (approval = 96.4%)
- $\delta_S = 0.01$: welfare = 84,833 (approval = 94.9%), a 28% decline
- $\delta_S = 0.02$: welfare = 52,104 (approval = 92.1%), a 56% decline

Under our welfare objective and sample composition, this finding shows that the Freddie-to-population default rate gap is a more consequential source of policy uncertainty than κ . Figure 2 compares the welfare and approval rate of each policy alternative.

Integrity gap. The LP relaxation ($z_c \in [0, 1]$) yields solutions that are nearly binary: at $\kappa = 0.10$ with standard tolerance, 0 out of 11,136 cells receive fractional values. With a tighter demographic parity tolerance ($\delta = 0.02$), only 2 cells are fractional; enumerating all $2^2 = 4$ binary completions of those fractional cells yields a welfare gap of 12 (0.01% of total welfare) relative to the LP upper bound. The continuous relaxation is thus a close approximation to the combinatorial problem.

5.3 Breakdown Grid

Table 9 reports the minimax-optimal approval rate across a grid of (κ, δ_S) values, quantifying the two-parameter sensitivity structure described in Section 4.2. The table confirms two patterns. First, the approval rate is insensitive to κ at any fixed δ_S : moving from $\kappa = 0$ to $\kappa = 0.20$ has negligible effect because the denied-applicant share is small relative to originated borrowers. Second, δ_S drives the reversal: the approval rate drops below the status quo (93.8%) between $\delta_S = 0.015$ and $\delta_S = 0.020$, regardless of κ . Figure 3 visualizes this contour.

5.4 Welfare Parameter Sensitivity

A natural concern is that the policy conclusions depend on the assumed welfare parameters. Table 10 reports the optimal approval rate across a grid of mortgage spread (r), loss given

default (LGD), and false-denial cost (FDC) values, holding $\kappa = 0.10$ and $\delta_S = 0$.

The “expand lending” conclusion is robust across most parameter combinations. It reverses only when LGD is high (0.50) *and* the false-denial cost is zero or very low, a calibration that ignores the social cost of credit rationing. Under the moderate benchmark ($r = 0.02$, $\text{LGD} = 0.35$, $\text{FDC} = 0.005$), the approval rate exceeds the status quo by 2.6 percentage points. Even at $\text{LGD} = 0.50$ with standard FDC, the optimal policy still approves 94–95% of applicants.

6 Conclusion

This paper provides partial identification bounds on the counterfactual default rate of denied mortgage applicants using linked HMDA and Freddie Mac administrative data, and conducts a two-parameter breakdown analysis over (κ, δ_S) . Our analysis yields three main conclusions.

First, policy conclusions are robust to substantial adverse selection. For any $\kappa \leq 0.10$ and transport gap $\delta_S \leq 0.01$, the minimax-optimal lending policy still approves more borrowers than the status quo. As a descriptive benchmark, the observed default gradient across denial-rate quintiles (2.2 percentage points) is roughly five times smaller than the κ values under which the “expand lending” conclusion holds.

Second, under our welfare objective and sample composition, transport uncertainty is a more consequential sensitivity dimension than adverse selection. This has a constructive implication: the most productive path to sharper policy conclusions is reducing the Freddie-to-population default rate gap through richer data linkage (e.g., incorporating performance data from multiple GSEs, servicers, or credit bureaus), rather than debating the plausibility of different κ values.

Third, lender screening contains real credit information (cells with higher denial rates exhibit monotonically higher observed default rates), but the informational content is modest relative to the variation in denial rates. The 3.5-fold increase in default rates across quintiles is substantially smaller than the 10-fold variation in denial rates, suggesting limited incremental screening precision at the margin.

Several limitations merit discussion. Our cell-level linkage is necessarily coarser than individual-level matching, potentially obscuring within-cell heterogeneity. The 25% of denied applicants in omitted cells (those without sufficient Freddie coverage) are subject only to uninformative Manski bounds. And the monotone selection assumption, while economically motivated, may be violated if lenders discriminate on dimensions correlated with risk.

Future research could extend this framework by incorporating additional data sources to tighten the transport assumption, developing data-driven methods for calibrating κ from

observable proxies, and applying the breakdown analysis approach to other credit markets where selective labels are prevalent.

Table 1: Summary Statistics: HMDA Mortgage Applications

Year	Apps	Originated	Denied	Den. Rate	Amt (\$K)	CLTV	Inc (\$K)
2018	2,607,789	2,384,344	223,445	8.6%	298	98.1	143
2019*	2,668,825	2,461,688	207,137	7.8%	310	83.8	124
2020	2,963,961	2,728,944	235,017	7.9%	332	85.0	127
2021	3,302,515	3,060,108	242,407	7.3%	378	82.8	232
2022	2,727,740	2,510,927	216,813	7.9%	407	90.2	158

Sample: conventional purchase, first-lien, owner-occupied, site-built, 1-unit. Action taken $\in \{1, 3\}$. Conforming loan amount restriction applied at cell construction (Section 3.3). [†]2018 CLTV uses single-lien LTV; 2019+ uses combined LTV. *2019 excluded from main specification due to COVID forbearance contamination of performance measures.

Table 2: Summary Statistics: Freddie Mac Loan Performance

Year	Loans	Credit Score	DTI	CLTV	90+ 24m	Severe 24m
2018	749,404	753	36.1	82.7	2.82%	0.014%
2019*	827,847	758	35.6	83.3	4.66%	0.004%
2020	969,719	759	34.7	83.6	1.79%	0.002%
2021	1,180,999	756	34.8	83.1	1.26%	0.002%
2022	841,467	755	37.0	81.8	1.58%	0.010%

90+ 24m = ever 90+ days past due within 24 months. Severe 24m = foreclosure, short sale, or REO within 24 months. *2019 contaminated by COVID forbearance reporting.

Table 3: Partial Identification Bounds: Denied-Group Counterfactual Default Rate

Sample	Cells	N_{apps}	N_{denied}	$\hat{p}$ (den-wt)	$[L, U] \kappa=0.05$	$[L, U] \kappa=0.10$	95% CI
Ex-2019*	11,136	9,998,525	623,090	0.0231	[.023, .073]	[.023, .123]	[.021, .126]
All years	13,745	12,245,770	758,694	0.0292	[.029, .079]	[.029, .129]	[.026, .133]
2018	2,507	2,187,741	149,602	0.0374	[.037, .087]	[.037, .137]	[.033, .142]
2020	2,752	2,584,111	153,574	0.0227	[.023, .073]	[.023, .123]	[.020, .125]
2021	2,971	2,871,041	164,802	0.0146	[.015, .065]	[.015, .115]	[.013, .116]
2022	2,906	2,355,632	155,112	0.0187	[.019, .069]	[.019, .119]	[.017, .120]
Full-sample**	33,098	13,193,704	1,014,718	—	[.022, .312]	[.022, .349]	—

Estimand: $E[Y(1)|D = 0, X \in c]$. Denied-weighted means. *Main specification. **Omitted cells use Manski $[0, 1]$. 95% CI: Imbens-Manski (2004), state $\times$ year cluster bootstrap ($B = 500$).

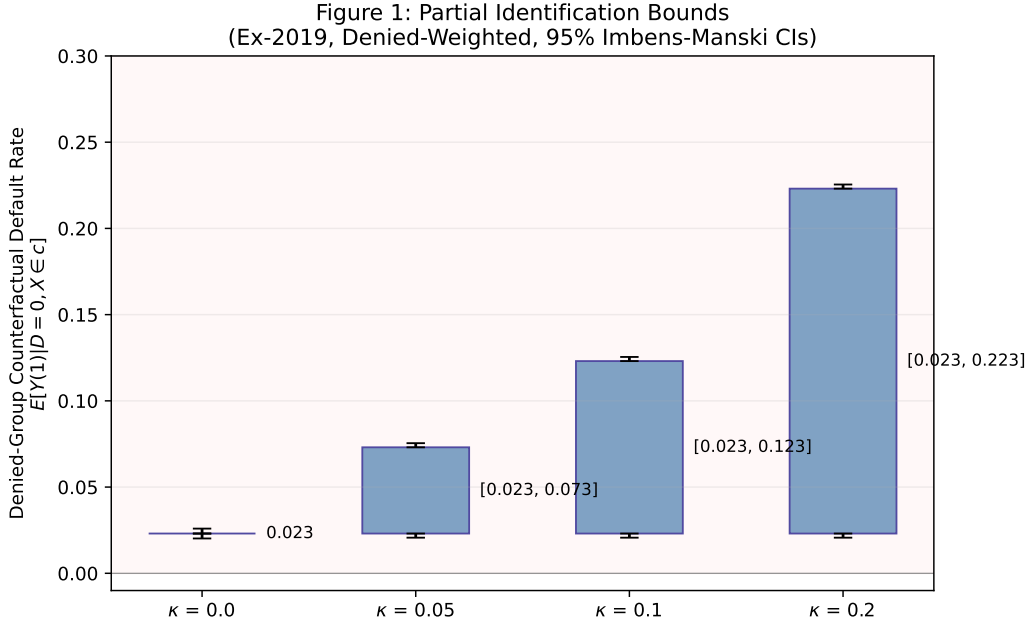


Figure 1: Partial Identification Bounds on Denied-Group Counterfactual Default Rate (Ex-2019). Each bar shows the κ -shift identified set $[\hat{p}, \hat{p} + \kappa]$ for $\kappa \in \{0, 0.05, 0.10, 0.20\}$, with whiskers indicating 95% Imbens-Manski confidence intervals. The shaded background marks the Manski no-assumption range $[0, 1]$. The width of each interval equals κ by construction.

Table 4: Heterogeneity by Cell-Level Denial Rate

Quintile	Denial Rate	N_{cells}	N_{apps}	N_{denied}	$\hat{p}$ (den-wt)	Upper ($\kappa=0.10$)
Q1 (lowest denial)	2.2%	2,228	1,572,253	34,869	0.0086	0.1086
Q2	3.4%	2,227	2,109,707	70,590	0.0120	0.1120
Q3	4.4%	2,227	2,452,624	108,521	0.0163	0.1163
Q4	6.1%	2,227	2,664,807	162,017	0.0242	0.1242
Q5 (highest denial)	20.6%	2,227	1,199,134	247,093	0.0305	0.1305

Cells grouped into equal-cell-count quintiles by denial rate (ex-2019, $\approx 2,227$ cells each). Q5 contains 40% of all denied applicants due to high per-cell denial counts. Denied-weighted bounds.

Table 5: Risk Characteristics Heterogeneity

Panel	Group	N_{cells}	Denial Rate	$\hat{p}$ (den-wt)	Upper ($\kappa=0.10$)
A: DTI	≤ 36	3,648	4.6%	0.0129	0.1129
	36–43	3,271	4.6%	0.0251	0.1251
	43–50	3,151	5.7%	0.0315	0.1315
	> 50	1,066	72.5%	0.0286	0.1286
B: CLTV	≤ 80	2,869	5.6%	0.0131	0.1131
	80–90	2,992	5.3%	0.0178	0.1178
	90–95	2,358	5.8%	0.0243	0.1243
	> 95	2,917	7.7%	0.0312	0.1312

Ex-2019, denied-weighted. DTI $> 50\%$ has extreme denial rate (72.5%) reflecting intensive screening.

Table 6: Transport Sensitivity ($\kappa = 0.10$, Ex-2019)

δ_S	Lower Bound	Upper Bound	Width
0.00	0.0231	0.1231	0.100
0.01	0.0146	0.1331	0.118
0.02	0.0087	0.1431	0.134
0.05	0.0020	0.1731	0.171

δ_S : transport band for $|p_A(c) - \hat{p}(c)|$.
Cell-level bounds: $[\max(0, \hat{p}(c) - \delta_S), \min(1, \hat{p}(c) + \delta_S + \kappa)]$. Reported values are denied-weighted aggregates across cells.

Table 7: Policy Learning Results (Ex-2019, $\kappa = 0.10$)

Policy	Approval Rate	Welfare
Status Quo (observed decisions) [†]	93.8%	126,415
Unconstrained ($\kappa = 0.10$)	96.4%	118,338
Transport-robust ($\delta_S = 0.01$)	94.9%	84,833
Transport-robust ($\delta_S = 0.02$)	92.1%	52,104
Binary (DP $\delta = 0.05$)	96.4%	118,338
Binary (DP $\delta = 0.02$)	97.2%	118,105

Welfare = $\sum_c z_c [n_c^{\text{orig}}(r - p_A^{\text{wc}} \cdot \text{LGD}) + n_c^{\text{den}}(r - p_R^{\text{wc}} \cdot \text{LGD})] - (1 - z_c)n_c \cdot \text{FDC}$. Parameters: $r = 0.02$, LGD = 0.35, FDC = 0.005. [†]Status quo uses observed $\hat{p}$ (not worst-case) and reflects within-cell screening; not directly comparable to cell-level robust policies.

Table 8: Coverage-Threshold Robustness ($\kappa = 0.10$, Ex-2019)

Threshold	Cells	Denied	Mean s	$\hat{p}$ (den-wt)	Bound $\kappa=0.10$
$s \geq 0$	11,136	623,090	0.50	0.0231	[0.023, 0.123]
$s \geq 0.2$	11,008	613,206	0.51	0.0231	[0.023, 0.123]
$s \geq 0.4$	5,512	328,754	0.68	0.0240	[0.024, 0.124]
$s \geq 0.6$	1,143	142,151	1.51	0.0281	[0.028, 0.128]

$s = n_{\text{Freddie}}/n_{\text{orig}}$: cell-level Freddie coverage ratio. Values of $s > 1$ can arise when Freddie purchases exceed HMDA-reported originations in a cell due to differences in reporting filters, purchaser coverage variation, or minor boundary mismatches in the cell-level linkage. Denied-weighted aggregate bounds.

Table 9: Breakdown Grid: Optimal Approval Rate over (κ, δ_S) (Ex-2019)

κ	δ_S					
	0.000	0.005	0.010	0.015	0.020	0.025
0.00	96.4	95.7	94.9	93.5	92.1	90.4
0.05	96.4	95.7	94.9	93.5	92.1	90.4
0.10	96.4	95.7	94.9	93.5	92.1	90.4
0.15	96.4	95.7	94.9	93.5	92.1	90.4
0.20	96.4	95.7	94.9	93.5	92.1	90.3

Optimal approval rate (%) from minimax welfare LP. $r = 0.02$, LGD = 0.35, FDC = 0.005. Status quo approval rate: 93.8%. Values above 93.8 imply the optimal policy expands lending relative to the status quo. The approval rate is nearly invariant to κ at each δ_S , confirming that adverse selection is not the binding dimension.

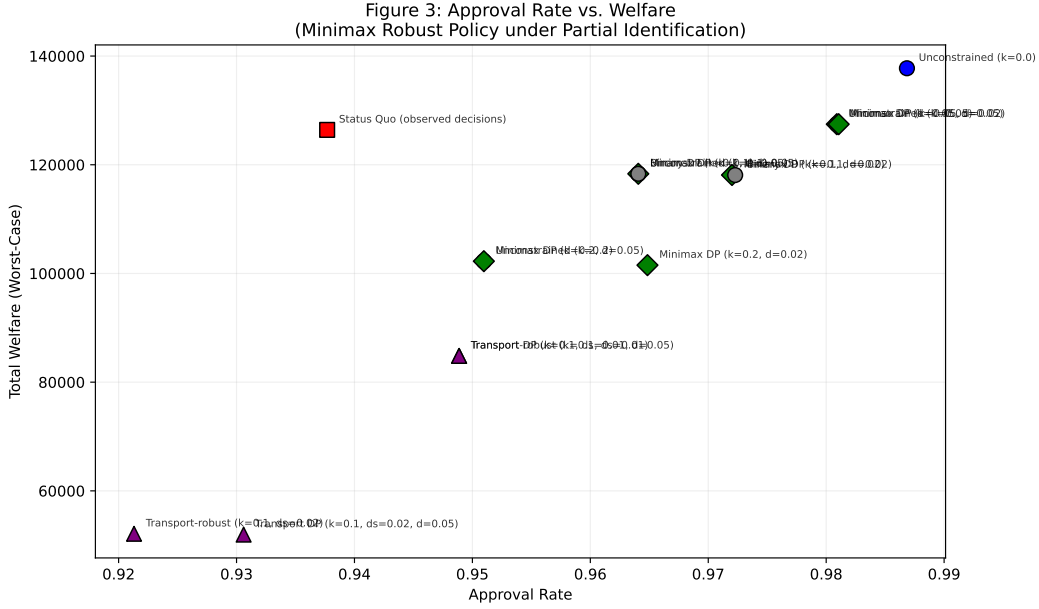


Figure 2: Policy Comparison: Approval Rate versus Worst-Case Welfare. Each point represents a policy alternative from the minimax welfare analysis, including the status quo, unconstrained policies at various κ , demographic-parity-constrained policies, and transport-robust variants, plotted by its optimal approval rate and total minimax welfare. Transport-robust policies shift left and downward as δ_S increases, illustrating that transport uncertainty is a more consequential source of welfare loss than adverse selection.

Table 10: Welfare Parameter Sensitivity ($\kappa = 0.10$, $\delta_S = 0$, Ex-2019)

r	LGD	Optimal Approval Rate (%)			Welfare	Δ vs SQ
		FDC = 0	FDC = 0.005	FDC = 0.01		
0.01	0.25	96.4	96.4	96.4	72,996	+2.6pp
0.01	0.35	96.4	96.4	96.4	55,447	+2.6pp
0.01	0.50	88.2	94.1	96.4	33,647	varies
0.02	0.25	96.4	96.4	96.4	144,463	+2.6pp
0.02	0.35	96.4	96.4	96.4	118,338	+2.6pp
0.02	0.50	93.1	95.2	96.4	82,654	varies
0.03	0.25	96.4	96.4	96.4	215,929	+2.6pp
0.03	0.35	96.4	96.4	96.4	181,228	+2.6pp
0.03	0.50	95.2	96.4	96.4	131,661	+2.6pp

Welfare and optimal approval rates under alternative parameter calibrations. r : mortgage spread; LGD: loss given default; FDC: false denial cost per applicant. Welfare is reported for the middle FDC column (0.005). The “expand lending” conclusion reverses only when LGD is high and FDC is zero or very low.

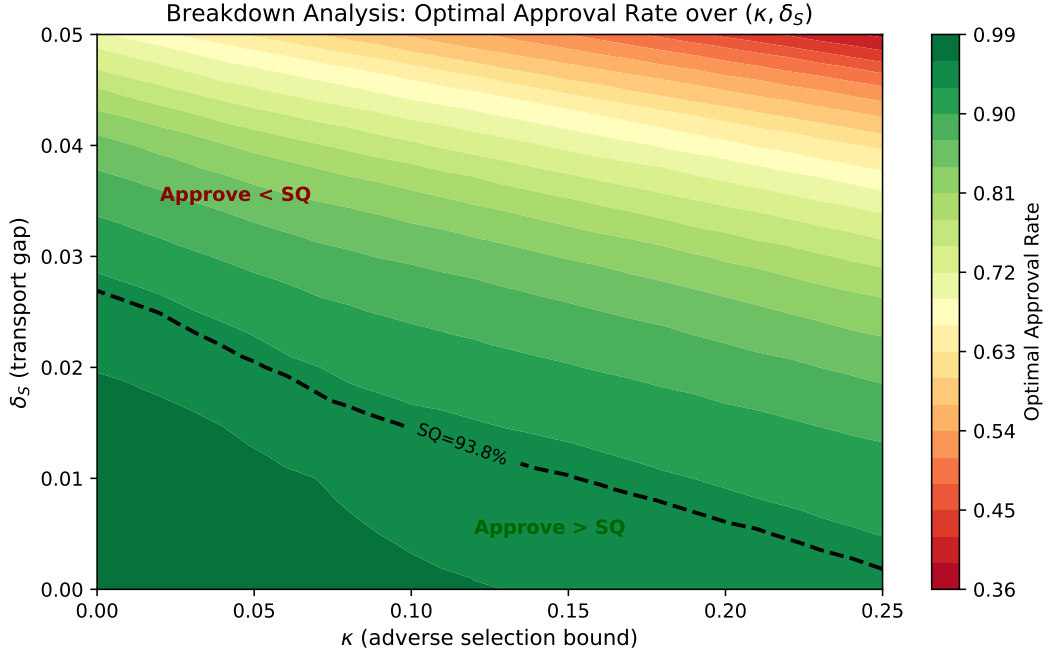


Figure 3: Breakdown Analysis: Optimal Approval Rate over (κ, δ_S) . The heatmap shows the minimax-optimal approval rate for each combination of adverse selection (κ) and transport uncertainty (δ_S). The black contour marks the status quo approval rate (93.8%). The approval rate varies primarily along the δ_S axis, confirming that transport uncertainty, not adverse selection, is the binding constraint on policy conclusions under our welfare objective and sample composition.

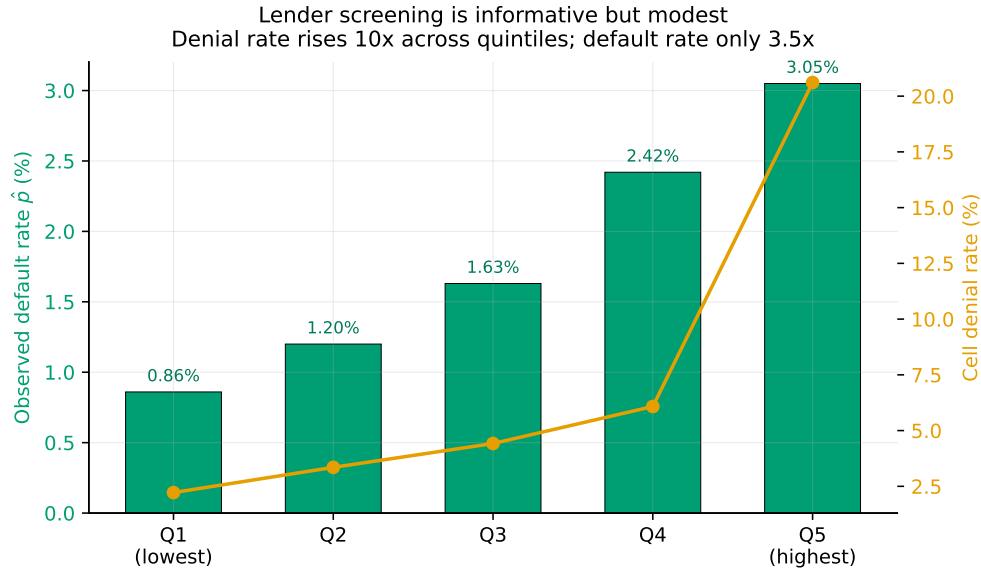


Figure 4: Lender Screening Gradient by Denial-Rate Quintile (Ex-2019). Bars (left axis) show the denied-weighted observed default rate $\hat{p}$ across cell denial-rate quintiles; the line (right axis) shows the cell denial rate. The denial rate rises about tenfold from Q1 to Q5, while the observed default rate rises only about 3.5-fold, indicating informative but modest screening at the margin of approval.

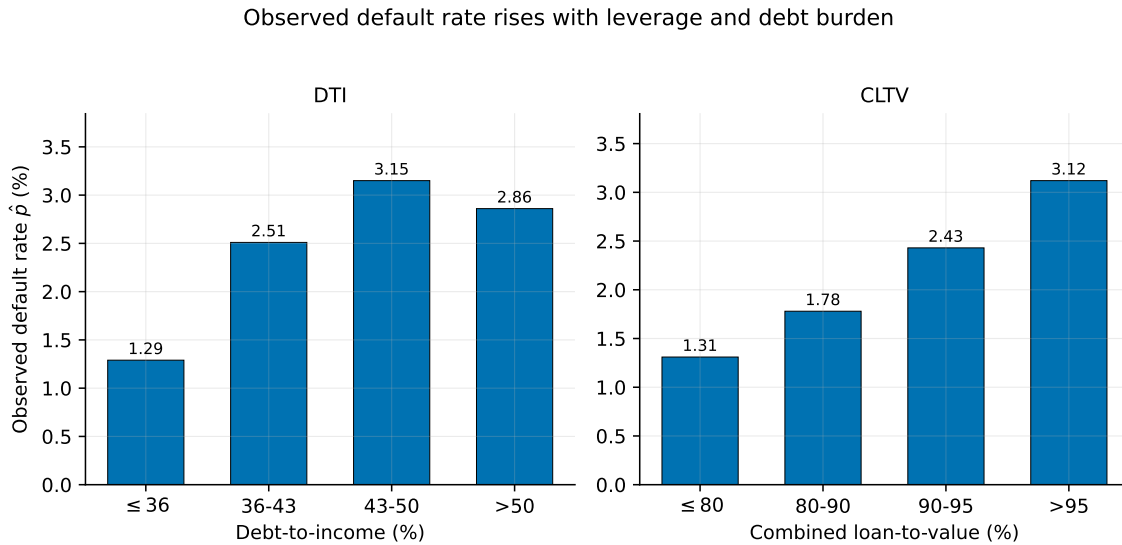


Figure 5: Observed Default Rate by Risk Characteristic (Ex-2019). Denied-weighted observed default rate $\hat{p}$ across debt-to-income (Panel A) and combined loan-to-value (Panel B) bins. The default rate rises with both leverage and debt burden, consistent with standard credit-risk gradients.

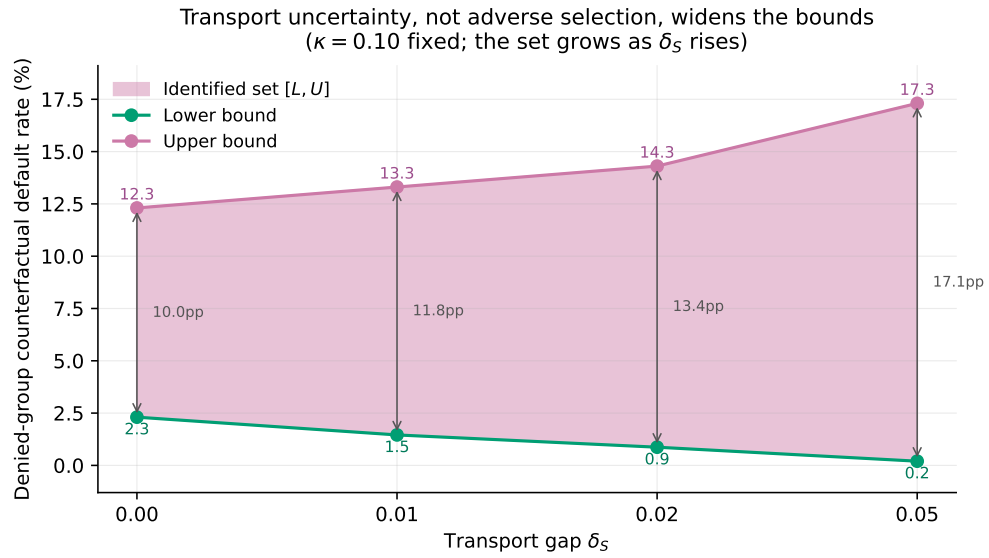


Figure 6: Transport Sensitivity of the Identified Set ($\kappa = 0.10$, Ex-2019). The shaded band is the denied-group counterfactual identified set $[L, U]$ as the transport gap δ_S increases. The set widens from 10.0 to 17.1 percentage points, confirming that transport uncertainty, rather than adverse selection, is the binding dimension under the welfare objective and sample composition.

A Geographic Heterogeneity

Table 11 reports bounds by Census region and by the eight states with the largest denied applicant volumes. Regional variation is modest: denied-weighted $\hat{p}$ ranges from 2.0% (Midwest) to 2.5% (Northeast). At the state level, Florida stands out with the highest $\hat{p}$ (3.1%) and denial rate (10.4%), while North Carolina has the lowest $\hat{p}$ (1.5%). These geographic patterns likely reflect variation in housing market conditions, borrower composition, and lender practices.

Table 11: Geographic Heterogeneity ($\kappa = 0.10$, Ex-2019)

Region/State	Cells	N_{denied}	Denial Rate	$\hat{p}$ (den-wt)	Upper	Mean s
<i>Census Region</i>						
Northeast	2,025	87,496	6.1%	0.0246	0.1246	0.49
Midwest	2,573	117,586	5.1%	0.0201	0.1201	0.49
South	3,843	288,997	7.3%	0.0242	0.1242	0.49
West	2,695	129,011	5.8%	0.0221	0.1221	0.54
<i>Top 8 States</i>						
FL	309	84,377	10.4%	0.0310	0.1310	0.65
TX	307	69,465	7.8%	0.0222	0.1222	0.60
CA	348	58,369	7.1%	0.0260	0.1260	0.67
NY	340	29,399	7.6%	0.0289	0.1289	0.53
IL	295	27,720	6.7%	0.0232	0.1232	0.72
GA	287	27,165	7.7%	0.0291	0.1291	0.60
NC	279	25,711	6.3%	0.0153	0.1153	0.47
MI	272	21,136	6.1%	0.0221	0.1221	0.52

Top panel: Census regions. Bottom panel: top 8 states by denied applicant volume. Denied-weighted aggregate bounds.

B Demographic Heterogeneity

As a supplementary analysis, we examine heterogeneity across borrower demographic groups using HMDA’s combined race-ethnicity variable. Table 12 and Figure 7 report denied-weighted bounds for the ex-2019 specification, restricted to cells with Freddie Mac coverage (matched support).

Point estimates of the observed default rate range from 2.1% (Asian) to 2.7% (NH Black, Hispanic), a spread of approximately 0.6 percentage points. Under $\kappa = 0.10$, all groups’ identification intervals overlap substantially: $[0.021, 0.121]$ for Asian, $[0.022, 0.122]$ for NH White, and $[0.027, 0.127]$ for NH Black and Hispanic. The modest cross-group variation in $\hat{p}$ should be interpreted with caution, as it reflects the race-blind within-cell performance

Table 12: Bounds by Race-Ethnicity ($\kappa = 0.10$, Ex-2019, Matched Support)

Group	N_{apps}	N_{denied}	Denial Rate	$\hat{p}$ (den-wt)	$[L, U]$	Coverage
NH White	6,360,614	298,959	4.7%	0.0216	[.022, .122]	92.9%
NH Black	499,916	65,052	13.0%	0.0273	[.027, .127]	92.3%
Hispanic	954,770	92,970	9.7%	0.0273	[.027, .127]	93.4%
Asian	777,309	58,767	7.6%	0.0209	[.021, .121]	94.9%

Combined race-ethnicity: Hispanic if derived ethnicity = Hispanic or Latino; otherwise NH White, NH Black, Asian. Within-cell race-blind performance: $\hat{p}$ common across groups within each cell. Coverage = share of group’s applications in matched cells. Bounds overlap substantially across all groups.

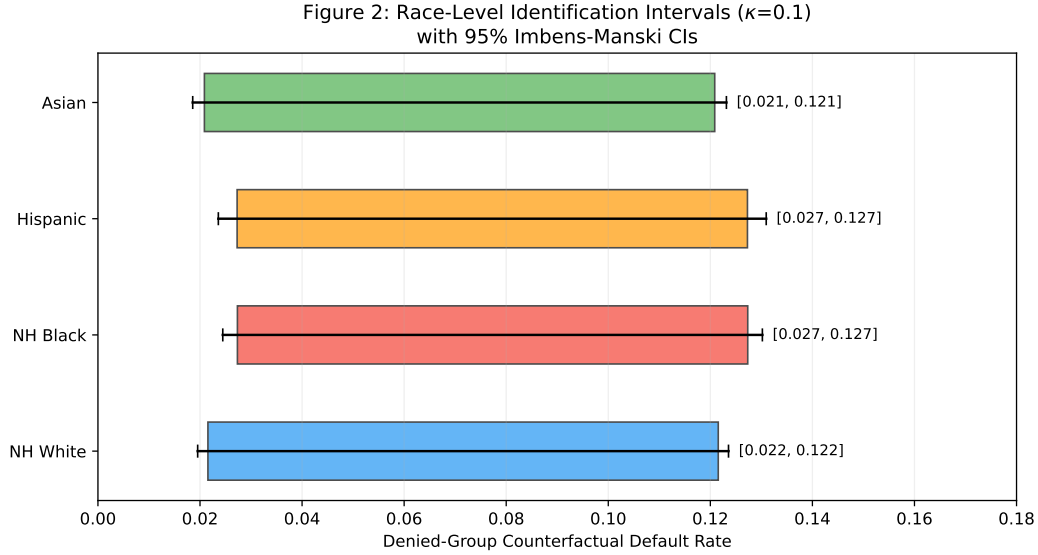


Figure 7: Partial Identification Bounds by Race-Ethnicity ($\kappa = 0.10$, Ex-2019). Each bar shows the identified set $[\hat{p}, \hat{p} + \kappa]$ for the denied-group counterfactual default rate. Bounds overlap substantially across all groups, reflecting the modest cross-group variation in cell-level observed default rates.

assumption: the Freddie Mac data do not contain race information, so race-specific $\hat{p}$ values are proxied by the cell-level rate. Relaxing this assumption via a race sensitivity parameter δ_R (allowing $|p_A(c, r) - \hat{p}(c)| \leq \delta_R$) further widens the group-specific bounds and renders cross-group comparisons uninformative at even modest values of δ_R .

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